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Mammalian Osteoderm Ultrastructure in the Armored *Acomys* Spiny Mouse Tail

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ABSTRACT

The skin of mammals acts as a barrier against pathogens, ultraviolet radiation, and mechanical trauma. Embedded within the dermal tail skin of spiny mice (genus *Acomys*) lie osteoderms—finely structured, symmetrically patterned, bone-like armor plates akin to those in armadillos and non-avian reptiles. Using 2D and 3D microscopic characterization techniques, we provide details on *Acomys* osteoderm patterning, cell size and distribution, and an ultrastructural overview of collagen-mineral relationships in the extracellular matrix. Osteoderms are organized into a periodic array with A-B-A-B-A-B patterning, with an angular offset to the central tail axis of roughly 26°, an aspect ratio of ~3.8, and a cellular mineral lacunar porosity of ~4.5%. Mature osteoderms in young mice preferentially have vascular channels in their distal-third portion, and marginally higher cellularity in their proximal portion. Patterned nucleation of nascent osteoderms occurs relative to surrounding guard and awl hairs, and develops through the growth and amalgamation of nanoscale mineral foci. On the epidermal-facing side of the osteoderm, bundled dermal collagen fibrils have prominent Sharpey's fiber insertions and a hypomineralized band along the osteoderm surface, whereas the dermis on the hypodermal side has parallel-oriented dermal collagen. This complex architecture may serve to define planes on which autotomy occurs.

1 | Introduction

The integumentary system – and primarily the skin – plays an important role in both physiological regulation and as a primary barrier that resists damage or stress from an animal's local environment (including mechanical abrasion, ultraviolet exposure, and pathogen ingress, amongst many others) [1]. As part of its structure-function relationship, the skin is most often subdivided into three principal layers: (i) the outermost epidermal layer, (ii) the central dermal layer, and (iii) the innermost hypodermal layer. Depending on their phylogeny, many organisms have developed specific adaptations to this base structure (e.g., through photoprotective hyperpigmentation [2], cornification or keratinization [3], or formation of chro-

matophores [4]) – creating fine variations in structural biology and gradual evolutionary adaptations to specific environments. Manifesting as the formation of bone-like mineralized armor plates in the dermis [5, 6], osteoderms are commonly found in extant non-avian reptile species (including crocodilians, lizards, turtles, and snakes – hereafter simply referred to as reptiles) as well as extinct Dinosauria, but they are uniquely distinct from the keratinized epidermis and calcified scales of fish [7]. Primarily composed of a calcium phosphate mineral in the organic extracellular matrix, osteoderms are theorized to provide a wide range of functions besides serving in organismal protection, including thermoregulation, hydrodynamic efficiency, cutaneous water retention in arid ecosystems, and general ornamentation [8].

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Among reptilian species, there is a wide range of osteoderm morphologies, encompassing imbricated plates, interlocking structures, and fused mosaics [9]. Heterogeneities in osteoderm consistency persist even at the level of a single animal, where osteoderms display inconsistencies in morphology and ubiquity depending on gross anatomical location [9–11]. At the microscale, reptilian osteoderms are described to have an interlaced network of dense-fibered collagen bundles that are intrinsically mineralized inside the osteoderm but unmineralized in the anchoring part [12]. A portion of the mineralized extracellular matrix in reptilian osteoderms reportedly does not take on the well-ordered structure of lamellar bone tissue but instead is more random in nature [12]. Reptilian osteoderm heterogeneity also persists in the microstructure/nanostructure of a single osteoderm [13], with clear evidence of elemental segregation [14, 15] and variation in mineral crystallite size/orientation [14].

In rare circumstances, osteoderms can be found in mammals (such as armadillos, or the extinct *Mylodontid* family of ground sloths). Recent work from Maden et al. [16] has found osteoderms to be present in the tail dermis of the extant *Deomyinae* *Acomys cahirinus* spiny mouse, and an associated upregulation of bone-forming genes such as *Osterix*, *Phex* (encoding for phosphate-regulating endopeptidase homolog X-linked), and *Ibsp* (encoding bone sialoprotein II production). Despite these spiny mouse osteoderms having developmental similarities to mammalian bone from a genetic standpoint, it is less clear from an ultrastructural perspective as only superficial electron microscopic analyses have been performed on mammalian osteoderms [17–19]. Here we provide a better alignment of the genetic understanding of osteoderms and known mineralized tissue structure. For this, we have used multiscale high-resolution 2D and 3D imaging protocols (particularly x-ray microscopy [XRM, submicron x-ray microcomputed tomography], focused ion beam scanning electron microscopy [FIB-SEM], and transmission electron microscopy [TEM]), to characterize the ultrastructure of mineralized osteoderms in the autotomic [20] *Acomys cahirinus* spiny mouse of the *Deomyinae* rodent subfamily – drawing comparisons to other osteoderms and known biomineralization systems.

2 | Results

2.1 | Histology of the *Acomys* Mouse Osteoderm

In the tail of newborn *Acomys cahirinus* spiny mice (1 day post-natal, P1), osteoderms are recently reported to both nucleate and grow within the dermal layer of the skin [16]. In a proximal region of the newborn *Acomys* tail, histology of transverse tail sections (stained with toluidine blue to highlight organic components in blue, and von Kossa reagent to highlight mineral components in black) shows that large, well-mineralized osteoderms indeed reside entirely within the dermis (Figure 1A). Although angularly offset from the central axis of the tail, osteoderms in these 2D sections are spaced below the stratum basale of the epidermis by at least 15 μm , and 150 μm from the upper boundary of the underlying hypodermis.

Sebaceous glands associated with prominent hair follicles are situated in the vicinity of the osteoderm. Guard hairs [21, 22] – observed to be the larger of the two hair types originating from the skin and typically forming the outer layer of the animal's coat – serve as a primary protective barrier for environmental interactions and an important means of thermal regulation. Awl hairs [21] – the smaller of the two hair types – form the undercoat and are flanked on either side by guard hairs (with an additional awl hair visible in the extradermal region). These triads of hair (two guard hairs and one awl hair) are nearly always centered with respect to the center of the osteoderm and are surrounded by sebaceous glands (containing sebum-producing sebocytes that maintain skin lubrication [23]). In this histology section, the sebaceous gland is separated from the osteoderm by roughly 40 μm . As the mouse grows, these measures may change, but their relative proportions could remain similar.

In the higher-magnification image of Figure 1B, although osteoblasts are not immediately visible nor abundant, cells consistent with an osteocyte phenotype are readily apparent within mineral lacunae located in the mineralized osteoderm. Because of the thinness of the osteoderm, most cells appear to be of the *osteoid osteocyte* cell type [24] where they are not completely encased within the mineralized matrix but instead have a partially unmineralized boundary at the osteoderm surface. This preferential form of exterior-facing ‘open’ porosity provides evidence for this being the major growth front side of the osteoderm. Also, at this epidermal-facing region, there also appears to be a higher abundance of unmineralized osteoid-like matrix relative to the hypodermal-facing portion of the osteoderm. This again provides evidence for a preferred site of collagen and mineral deposition on the exterior surface of the osteoderm and suggests that osteoderms nucleate deeper in the dermis (then predominantly grow toward the epidermis).

Of note in the osteoderms is the frequent orientation of bundled dermal collagen fibrils on both the epidermal-facing and hypodermal-facing regions of the osteoderm. The osteoderm is roughly positioned at the junction between the papillary dermis (the outer dermal layer) and reticular dermis (the inner dermal layer). In the papillary dermis, the dermal collagen bundles are oriented mostly oblique to the osteoderm surface – acting as so-called Sharpey's fibers to help anchor the osteoderm within the surrounding dermal tissue, perhaps in a manner analogous to other mineralized tissues including bone and tooth cementum [25–28]. The same observation is less apparent on the inner surface of the osteoderm, where dermal collagen bundles in the reticular dermis instead lie in clear planes that are roughly parallel to the surface of the osteoderm (without a clear oblique orientation or insertion). Thus, the anchorage of the osteoderm seems to be more heavily influenced by its interaction with the papillary dermis rather than the reticular dermis.

When sections are stained with only toluidine blue (Figure 1C), there appears to be some peripheral staining that surrounds the entire osteoderm along with individual cell lacunae – delimiting unmineralized matrix in these areas. The approximate 20 μm space between neighboring osteoderms, as in Figure 1D,E) tends to be filled with either connective tissue or fine vasculature.

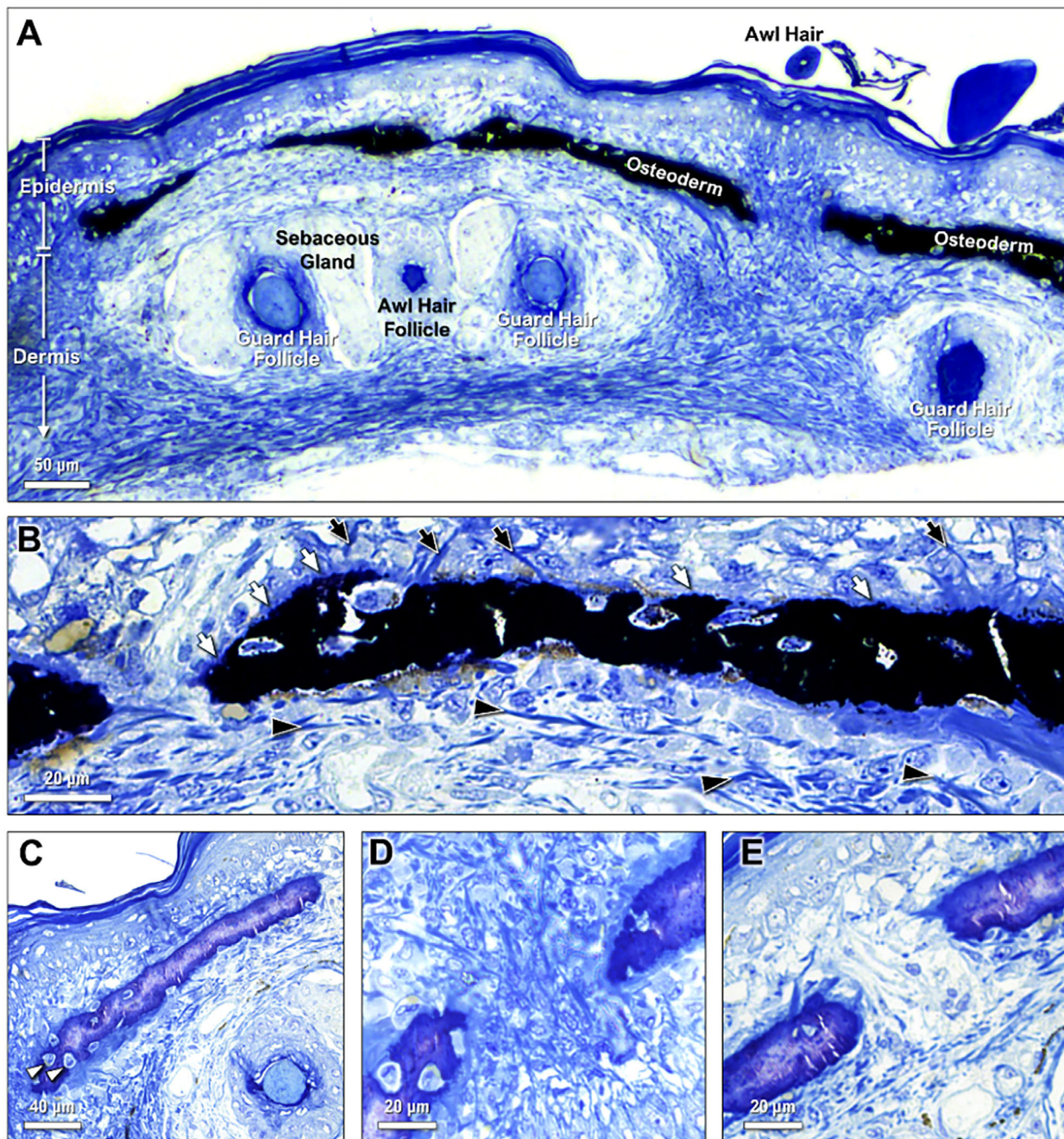


FIGURE 1 | Light microscope histology of the *Acomys cahirinus* spiny mouse osteoderm. (A) von Kossa staining (black, for mineral components) and toluidine blue (for organic components) show the position of the osteoderm within the dermis of the tail. Note the position of hair follicles relative to the osteoderms. (B) A multitude of collagen arrangements can be found surrounding the osteoderm, including fine osteoid-like collagen (white arrows), bundled dermal collagen with a Sharpey's fiber orientation in the papillary dermis (black arrows), and dermal collagen with a parallel orientation to the osteoderm in the reticular dermis (black arrowheads). (C) Toluidine blue staining alone of the osteoderm appears more intense surrounding the cell lacunae (white arrowheads) and at the periphery of the osteoderm. (D,E) The space between neighboring osteoderms is often filled with dense connective tissue or vasculature.

2.2 | Comparing the Morphology of Well-Developed and Developing Osteoderms

To visualize long-range order of three-dimensional osteoderm morphology in the spiny mouse tail, X-ray microscopy provides an excellent balance between a large field of view and high-resolution imaging. A stitched and reconstructed X-ray

microscopy volume of an entire newborn *Acomys* tail can be found in Figure 2A and Video S1. The distribution of the osteoderms is highly ordered around the vertebrae, taking on an A-B-A-B-A-B repeating pattern (where the proximal-to-distal midline of one osteoderm is aligned with the gap between two osteoderms in the subsequent concentric ring) along the entire length of the tail. Deviations in this repeating structure (akin to

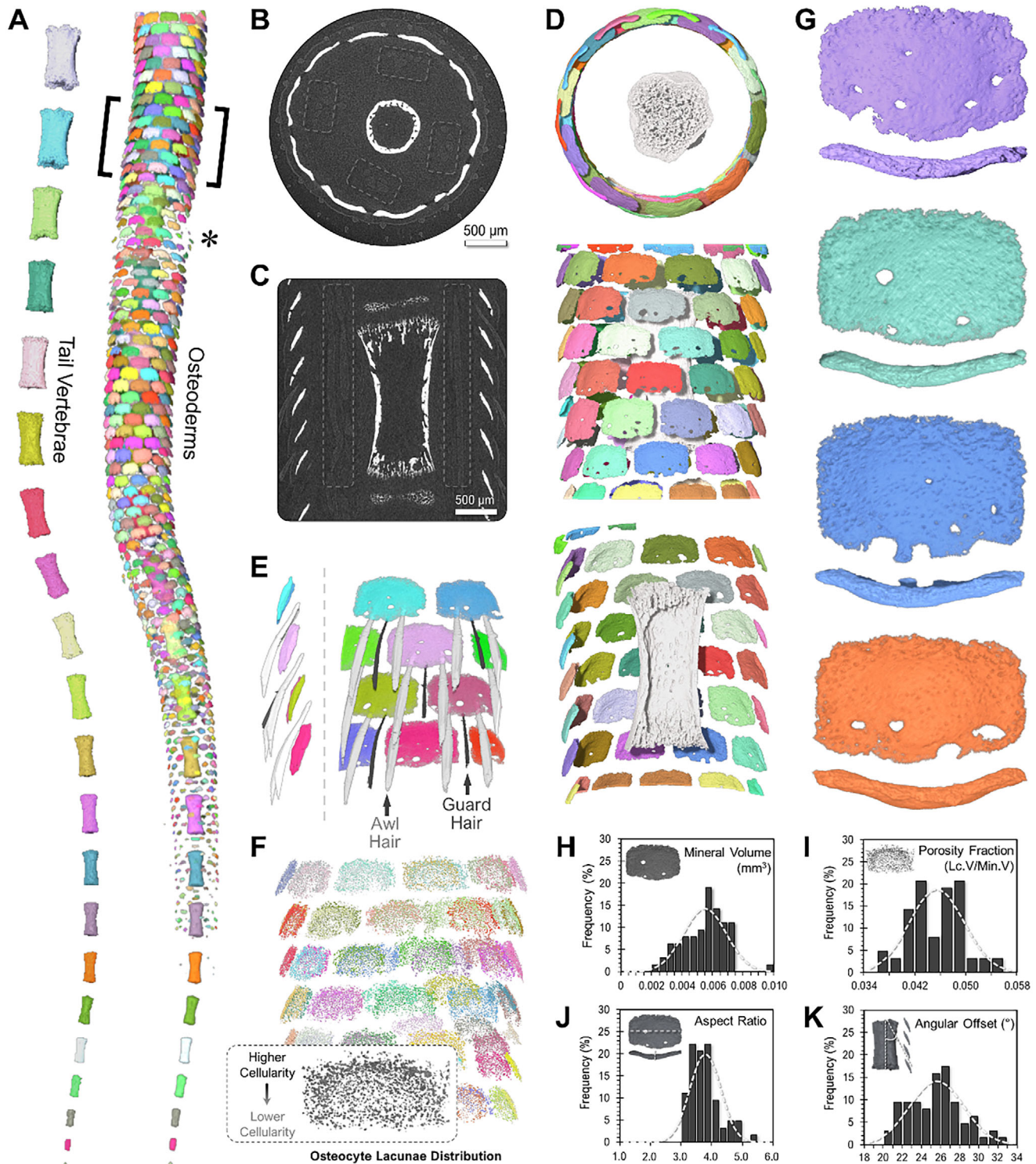


FIGURE 2 | X-ray microscopy of the most mature osteoderms in the proximal tail of a newborn (P1) spiny mouse. (A) Visualizations of the young *Acomys cahirinus* tail showing segmented tail vertebrae (with and without digital removal of osteoderms). The asterisk represents a region of active regeneration following tail damage, and the bracketed region is magnified in subsequent panels. (B,C) 2D reconstructed cross-sectional and longitudinal planes of osteoderms in the proximal tail surrounding a centrally located vertebra. Osteoderms have a clear concentric order and are slightly angled relative to the vertebral long axis. Tail tendon is faintly visible in the dashed boxes. (D) Higher magnification renderings showing segmentation of a tail vertebra and surrounding osteoderms in front profile, transverse cross-sectional, and longitudinal cut planes. (E) Front and side profiles show the distribution of mature osteoderms in relation to awl and guard hairs. Hairs are interposed between concentric osteoderm rings. (F) Segmentations of mineral lacunar porosity in the individual osteoderms. Lacunar porosity appears marginally higher in the proximal region of the osteoderm. (G) High-magnification frontal and proximal edge-on profiles of four separate osteoderms. Individual osteoderms are slightly curved and have an abundance of large vascular canals located in their distal half. (H–K) Quantitation of mineral volume, mineral lacunar porosity fraction, aspect ratio, and angular offset of mature osteoderms. See also: Video S1.

'edge dislocations' in crystallography) are rare, but do occur in select locations along the tail (Figure S1). In a damaged region of the tail (where osteoderms are fractured and/or are in the process of repair), osteoderms appear to form in the exact location of the prior osteoderms – propagating the same imbricated A-B-A-B-A-B pattern despite the initial disruption in tail integrity. The stitched scan of the osteoderm also shows a net decrease in mineralized osteoderm volume in the distalmost tail region of these young mice. Generally, there are ten osteoderms within each well-developed concentric ring.

Higher-magnification x-ray microscopy in a proximal region of the tail is shown in Figure 2B–G and Video S1. A transverse section through a single concentric ring of osteoderms and tail vertebra (Figure 2B) shows the location of the osteoderms with respect to the vertebra. The outer layer of the epidermis is also faintly visible beyond the exterior of the osteoderms, and four groups of tail tendon can be seen situated between the vertebra and osteoderms. In a reconstructed longitudinal plane through the tail (Figure 2C), the osteoderms are seen to have a consistent degree of inclination from the central vertebral axis (where the proximal portion of the osteoderm is located closer to the vertebra, and the distal portion of the osteoderm is more distant from the vertebra). Secondary ossification centers are also visible proximal and distal to the vertebra in this longitudinal view. 3D segmentations of the major mineralized components (vertebral bodies and osteoderms) are shown in Figure 2D, where the short-order osteoderm array forms a near-perfect imbricated cylinder and the vertebral body is slightly off-center from the cylindrical axis. In 3D, the triad of two awl hairs and one guard hair is clearly linked to the position of the osteoderm (Figure 2E) and is also highly periodic. The hairs follow the same angular offset to the vertebral axis as the osteoderms where they are sandwiched between concentric osteoderm bands, although they extend well beyond the epidermal layer. For the mature osteoderms, the corresponding guard hair is situated directly beneath the proximal-to-distal midline of the osteoderm in a highly consistent pattern. The two associated awl hairs also lie beneath the osteoderm (flanking the guard hair on either side) and are spaced from the guard hair by approximately 250–350 μm .

The fine mineral lacunar porosity of each osteoderm (Figure 2F) appears elevated in the proximal portion of each osteoderm. The mineral lacunar volume fraction (measured as Lc.V/Min.V) for the proximal half of the well-developed osteoderms was calculated to be $5.03 \pm 0.48\%$, while the mineral lacunar volume fraction in the distal end was less at $4.04 \pm 0.50\%$ (yielding $p < 0.001$ in an independent t-test). All twenty osteoderms used for these mineral lacunar porosity measurements had a higher lacunar volume fraction at their proximal end. Regardless of whether a proximal or distal region of the osteoderm was examined, the lacunar porosity values in the osteoderm are both substantially higher than measurements obtained from human cortical bone (which has a Lc.V/Min.V of 0.84% when measured with synchrotron radiation using a similar voxel size [29]) – suggesting an unusually high degree of cellularity in these mammalian osteoderms. En face views from representative well-developed osteoderms can be seen in Figure 2G, where the osteoderm has a slight curvature that follows the shape of the tail, and the surface-connected mineral lacunar porosity is visible as fine dimples across the osteoderm. Larger vascular channels

also permeate through the osteoderm (with a roughly cylindrical appearance, or hemicylindrical if at the osteoderm periphery). These large vascular channels tend to be preferentially distributed toward the distal edge of the osteoderm and rarely appear in the most proximal third.

Quantification of mature osteoderm characteristics from 61 osteoderms can be found in Figure 2H–K. In neonatal spiny mice, the well-developed osteoderms have an average size of $627 \times 416 \times 114 \mu\text{m}$, an average mineralized volume of $0.0055 \pm 0.0014 \text{ mm}^3$, an average lacunar porosity (Lc.V/Min.V) of $4.54\% \pm 0.43\%$, an average aspect ratio of 3.78 ± 0.50 , and an average angular offset to the vertebral long axis of $25.6^\circ \pm 2.8^\circ$. This angular offset contributes to their mechanical function through overlap and permits intervening hair follicles to span the dermis and epidermis. Mechanical functions and a potential role in prey survival remain to be defined.

Toward the distal end of the tail (Figure 3A), early-stage osteoderms are in development at this young age. High-resolution XRM analysis (Figure 3B–D; S1) shows the distribution of nascent osteoderms in the developing region. Here, osteoderms are much smaller than in the proximal tail, but are still uniquely positioned into concentric rings along transverse sections. In these concentric rings, there are typically only eight osteoderms (fewer than the ten present in the well-developed tail) – likely proportional to the tail diameter in this region. As was seen for the well-developed osteoderms, the vertebral body is slightly off-center relative to the cylindrical arrangement of developing osteoderms. Since vascular channels are frequently seen in the newly forming osteoderms (but only in the distal fraction of fully developed osteoderms), it stands to follow that the nucleation of an osteoderm begins in its distal half with mineralization proceeding in a distal-to-proximal direction.

The hair follicles in this partially armored region of the tail (Figure 3E) are arranged into well-aligned columns of larger awl hairs or smaller guard hairs, in a pattern that appears consistent with the triad of hairs surrounding each well-developed osteoderm. Despite slightly varying morphology, each of the developing osteoderms also appears highly cellular (Figure 3F), with a number of small, surface-connected lacunae visible in the mineralized regions that house cells. Mapping the size gradient of osteoderms along the distal region of this particular tail (Figure 3G) shows a sigmoidal distribution Equation (1). Distal osteoderms (nearest the most distal vertebra) show a near-exponential size gradient in terms of mineralized volume across the first $\sim 10 \text{ mm}$ of nucleated osteoderms, followed by $\sim 10 \text{ mm}$ of linear size gradient that then plateaus toward the proximal region of the tail (although the largest overall osteoderm is indeed located in the mid-tail in this specimen). A small region of tail damage disrupts the size gradient trend in this tail, where osteoderms are smaller due to fracture and/or regeneration in this region.

$$V [\mu\text{m}^3] = \frac{2.6 \times 10^7}{1 + e^{-0.33(x_{[\text{mm}]} - 25.3)}} \quad (1)$$

A quantitative characterization of developing osteoderm characteristics can be found in Figure 3H–J. Measured across the 52 distal-most osteoderms, the average mineral volume is $140,912 \pm$

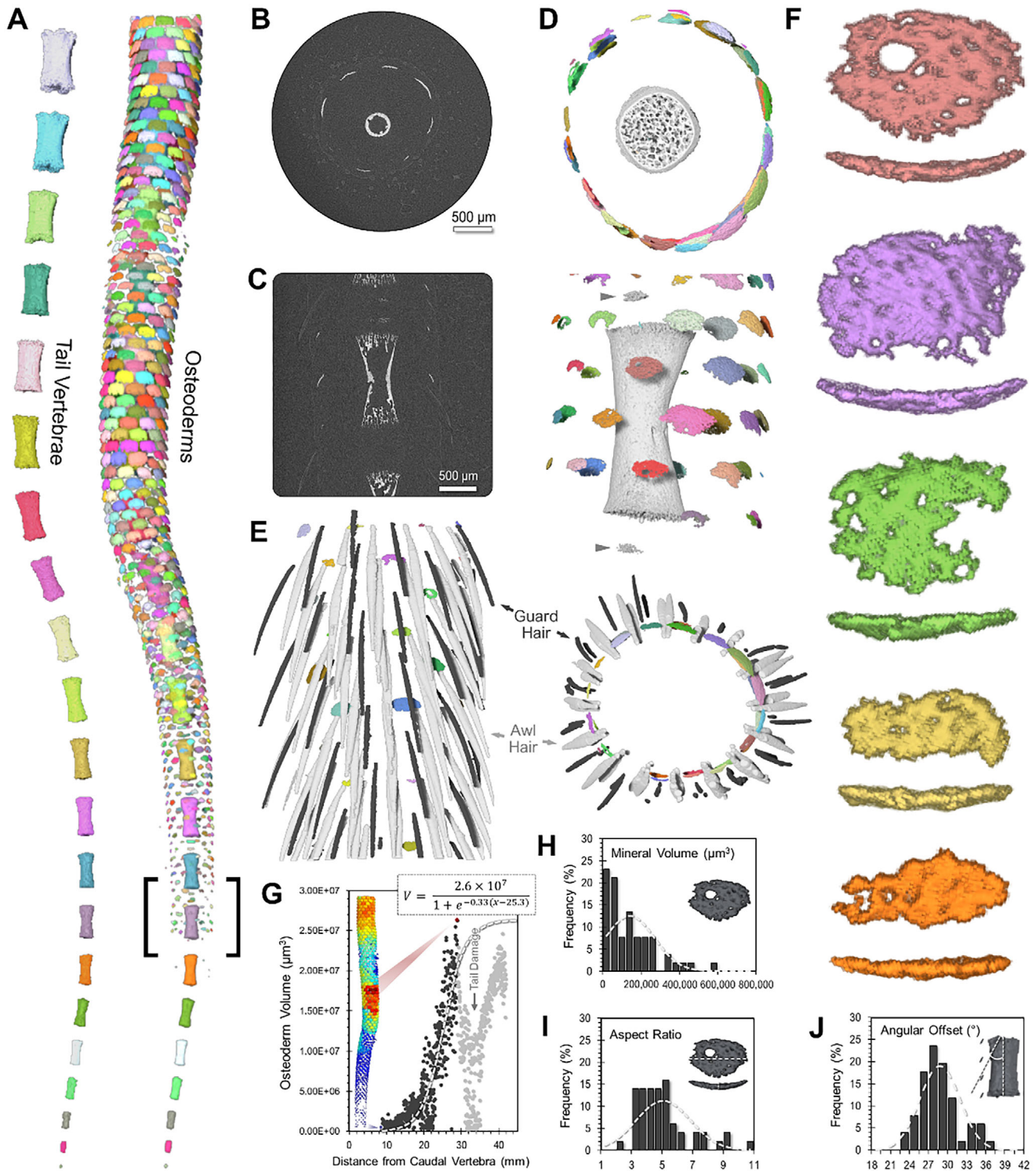


FIGURE 3 | X-ray microscopy of developing osteoderms in the distal tail of a newborn (P1) spiny mouse. (A) Visualizations of the young *Acomys cahirinus* tail showing segmented tail vertebrae (with and without digital removal of osteoderms). The bracketed region is magnified in subsequent panels. (B,C) 2D reconstructed planes of actively developing osteoderms in the distal tail. Osteoderms maintain their concentric positioning and angular offset to the vertebral long axis. (D,E) Higher magnification renderings showing segmentation of the tail vertebrae, osteoderms and awl/guard hairs. Despite being smaller, the osteoderms maintain their relative position to the hairs and their repeated A-B-A-B-A-B pattern. (F) High-magnification frontal and proximal on-edge profiles of five separate osteoderms. Developing osteoderms frequently have vascular canals embedded within, suggesting that they nucleate in their distal region and grow proximally. (G) Correlation between osteoderm size and location in the tail. Osteoderms grow in a size gradient that spans roughly 20 mm before reaching maturity. (H–J) Quantitation of mineral volume, aspect ratio, and angular offset in developing osteoderms. See also: Video [S1](#).

127,448 μm^3 and the average aspect ratio is 5.07 ± 1.77 . Inherently, these developing osteoderms are much more variable in morphology than their well-developed counterparts. The higher aspect ratio may indicate a slightly preferential rate of initial growth in the circumferential direction (as opposed to the proximal-to-distal direction), but there is no clear trend that persists between dorsal, ventral, leftward, or rightward osteoderms. The average inclination angle of the developing osteoderms is $28.8^\circ \pm 3.1^\circ$ – perhaps at a slightly greater angle to the vertebra than in the proximal tail.

2.3 | Sharpey's Fibers and Mesoscale-to-Nanoscale Osteoderm Mineralization

Scanning electron microscopy with backscatter imaging of a well-developed osteoderm is shown in Figure 4 following specimen osmication, serial dehydration, epoxy embedding, and mechanical polishing of the sample surface, with Figure 4A showing an example of a mature osteoderm. The hypodermal-facing side of the osteoderm has a sharp and continuous boundary separating the mineral and organic phases, with little evidence of ongoing mineralization. The epidermal-facing side of the osteoderm, however, is marked by a trilayered structure, consisting of: i) the bulk of the densely mineralized osteoderm, ii) a hypomineralized shell, with a thickness of $\sim 1 \mu\text{m}$, and iii) a thin, but densely mineralized periphery with a thickness of $\sim 200 \text{ nm}$. The trilayered structure is present along the entirety of the epidermal-facing facet of the osteoderm. The hypomineralized matrix is much more ubiquitous in regions of active osteoderm mineralization (Figure 4B), with energy-dispersive x-ray spectroscopy (EDS) confirming the presence of calcium phosphate material within in the organic extracellular matrix phase (nitrogen-rich) in the actively mineralizing region of the osteoderm. Incipient mineral foci (Figure 4C,D) form both within this hypomineralized band and directly in the dermal collagen on either side of the osteoderm in a process that resembles classical nucleation and growth. Following growth and impingement of these mineral foci, the mineral phase takes on a monolithic morphology with a disordered network of organic inclusions rather than having the clear crossfibrillar mineral tessellation pattern found in lamellar bone [30].

Insets of mineral foci in the osteoid collagen (Figure 4C) and the hypomineralized/active band of the osteoderm (Figure 4D) both show the presence of small, rounded granules (~ 50 – 200 nm in size) with a higher backscatter electron yield than the surrounding material. Multiple granules are often encapsulated within individual mineral foci. These isolated granules are not typically found in the monolithic mineral of the osteoderm, but there are definitive clusters of round, organic, and hypomineralized inclusions encapsulated within the bulk mineral (Figure 4E) that contain these granules. In 2D images, these inclusions are grouped in clusters of ~ 10 – 30 and are more densely found near the hypodermal-facing facet of the osteoderm.

Focused ion beam – scanning electron microscopy (FIB-SEM) tomography was used to generate high-resolution, 3D reconstructions of the mineralizing osteoderm across a $37.4 \times 46.4 \times 66.3 \mu\text{m}$ imaging volume. The resulting segmentation of mineral can be found in Figure 5A and Video S2, where regions of coalesced and

coalescing mineral foci co-exist in the tissue. $10 \times 10 \times 10 \mu\text{m}$ insets of each morphology can be seen where regions containing coalesced/impinged mineral foci present with a relatively flat/planar surface, presumably stemming from a more passive mineralization state. A few isolated mineral foci can be found surrounding the flatter surface of the osteoderm, but these are sparsely distributed. In a nearby site of active mineralization, rounded mineral foci are seemingly actively accreting additional calcium and phosphate as a precursor to the coalesced state – as occurs in other mineralized tissues [31]. Mineral contained in this active sub-region has a much higher surface area-to-volume ratio than the densely mineralized region nearby, with individual foci in the active region occupying only 21% of the total $10 \times 10 \times 10 \mu\text{m}$ volume.

A reconstructed plane of the FIB-SEM imaging volume shows the difference in cellularity between the epidermal- and hypodermal-facing facets of the osteoderm (Figure 5B). On the epidermal side, cell lacunae are abundant and house cells consistent with being osteoid osteocytes [24] based on their location and surface connectivity. Intervening spicules of mineralized tissue separate these osteoid osteocytes, sometimes by as little as $3 \mu\text{m}$. Large bundles of dermal collagen fibrils can be seen in the organic matrix on either side of the osteoderm, with additional cells (likely fibroblasts) dispersed throughout. Segmentations of mineral, cell nuclei, and fibroblast nuclei show the approximate cell density in the surrounding matrix (Figure 5C; Video S2). On the epidermal side, large fibril bundles in the dermal collagen are obliquely inserted into the hypomineralized band of the osteoderm – forming an arrangement of Sharpey's fibers (Figure 5D; Video S2) that are absent on the hypodermal-facing portion of the osteoderm (Figure 5E; Video S2), as consistent with the histology shown previously. The Sharpey's fiber insertion is along the same surface as the hypomineralized trilayer on the epidermal-facing facet osteoderm, where the hypomineralized trilayer contours in 3D to follow the shape of the mineral lacunae (Figure 5F).

Transmission electron microscopy at various sites in and around a well-developed osteoderm is shown in Figure 6. Selected-area electron diffraction reveals that the mineral phase of the osteoderm is consistent with being a polycrystalline hydroxyapatite phase, with diffraction visible from both the (002) and (004) crystallographic planes (Figure 6A). Based on the large arc length for (002) diffraction, individual apatite crystals are assumed to be distributed without major preferred orientation of their *c*-axes. Type I collagen fibrils in the surrounding osteoid are loosely organized into twisted arrays that run roughly perpendicular to the osteoderm surface to interact with this mineralized collagen of the osteoderm (Figure 6B).

In the dermis (far outside of the osteoderm, Figure 6C), type I collagen fibrils are instead organized into tight bundles that can be up to $2 \mu\text{m}$ in diameter. Individual fibrils within these bundles are highly unidirectional, but different bundles exhibit differences in their collagen orientation. The dermal collagen is rich in filamentous proteoglycans (Figure 6D, ~ 1 – 5 nm in diameter) – especially visible bridging less-bundled collagen fibrils in the micrograph. The osteoderms mineralize in such a way that they inevitably encapsulate select bundles of dermal collagen to become insertions (visible as the higher-contrast areas inside the osteoderm in Figure 6A). A higher-magnification

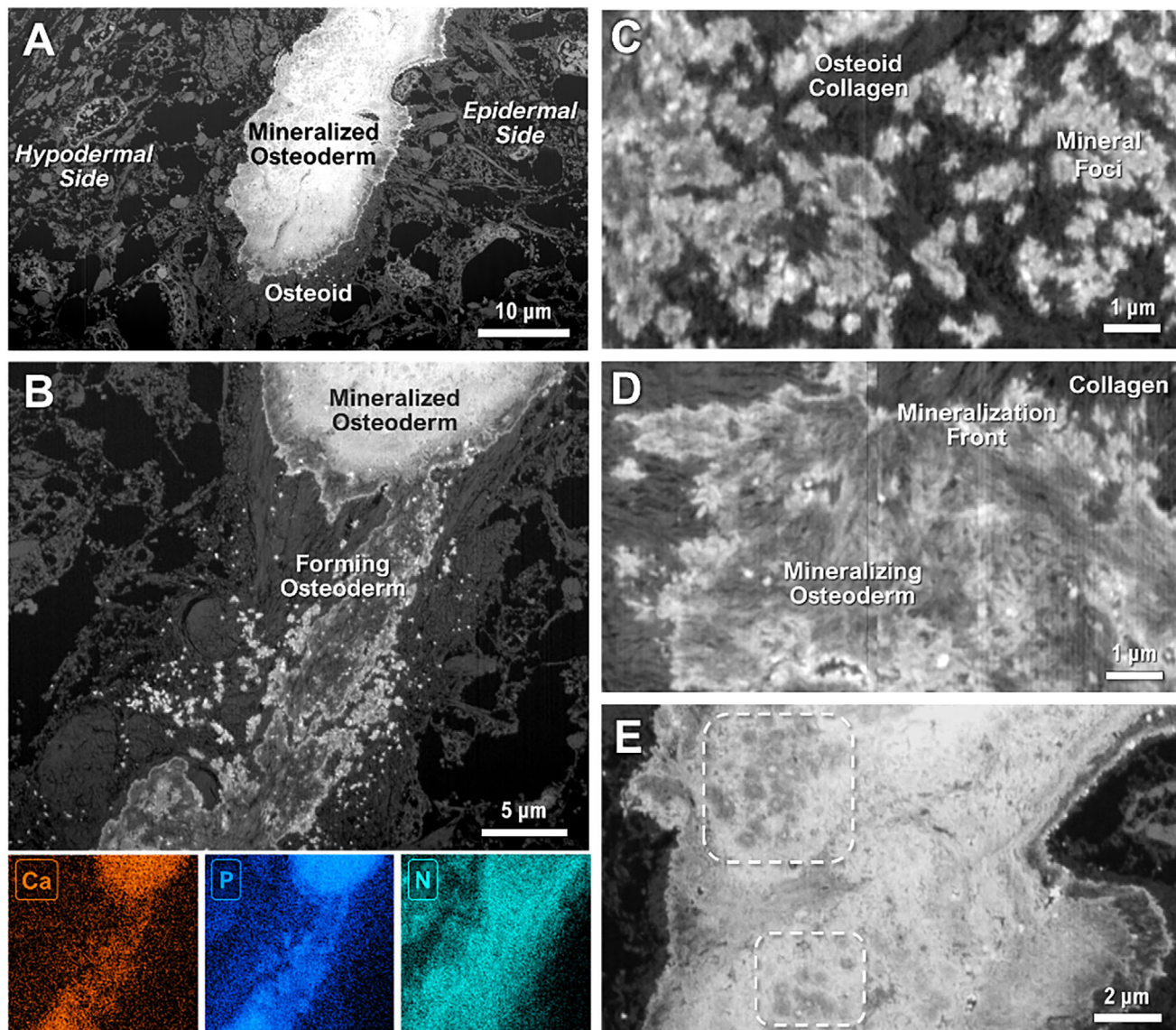


FIGURE 4 | Scanning electron microscopy of osteoderm structure in the proximal tail of the spiny mouse. (A) The mineralized osteoderm in this region is roughly 15 µm in thickness, with osmication enhancing the contrast of the surrounding soft tissue. The epidermal-facing surface of the osteoderm contains a hypomineralized trilateral of seemingly ongoing mineralization, which is absent on the hypodermal-facing surface. (B) In a region of apparent active mineralization, nanoscale incipient mineral foci can be found inside a wide hypomineralized band. Energy-dispersive X-ray spectroscopy confirms the presence of abundant calcium and phosphorus throughout this region. (C,D) High-contrast, rounded mineral granules (~ 50–200 nm in diameter) can often be found at the center of each individual mineral focus – both within the hypomineralized band and in otherwise mineralizing osteoid. (E) View of the osteoderm in cross-section showing clusters of round organic inclusions (dashed boxes) in the hypodermis-facing region. These have a densely mineralized core, similar to the round mineral granules in nucleating mineral foci.

view of these bundled dermal collagen fibrils can be seen in Figure 6E, where the collagen lies in the plane of the image and displays a clear banding pattern typical of fibrillar collagens. A higher abundance of apatitic platelets can be found in the disordered osteoid collagen fibrils at the periphery of the osteoderm (Figure 6F), where mineralization does appear to span multiple fibrils (presumably through both extrafibrillar and intrafibrillar means of mineralization). Individual crystals of apatite mineral can be seen penetrating the transverse (circular) cross-sections of the individual osteoid fibrils, and also propagating along the surface of nearby longitudinally oriented osteoid fibrils.

TEM also shows that the globular organic inclusions can be found inside discretely mineralized domains in the actively mineralizing osteoid of the osteoderm (Figures 6G,H). Individual mineral crystals are visible around the periphery of each mineralized domain, giving rise to a ‘feathered’ exterior texture. The globular inclusions themselves are of unknown composition but consist of an electron-dense outer shell with a high-contrast interior. The collagen in this region is loosely unidirectional, similar to that of the Sharpey’s fibers.

To view the extent to which the bundled dermal collagen is Sharpey’s fibered, bundles of dermal collagen on the exterior

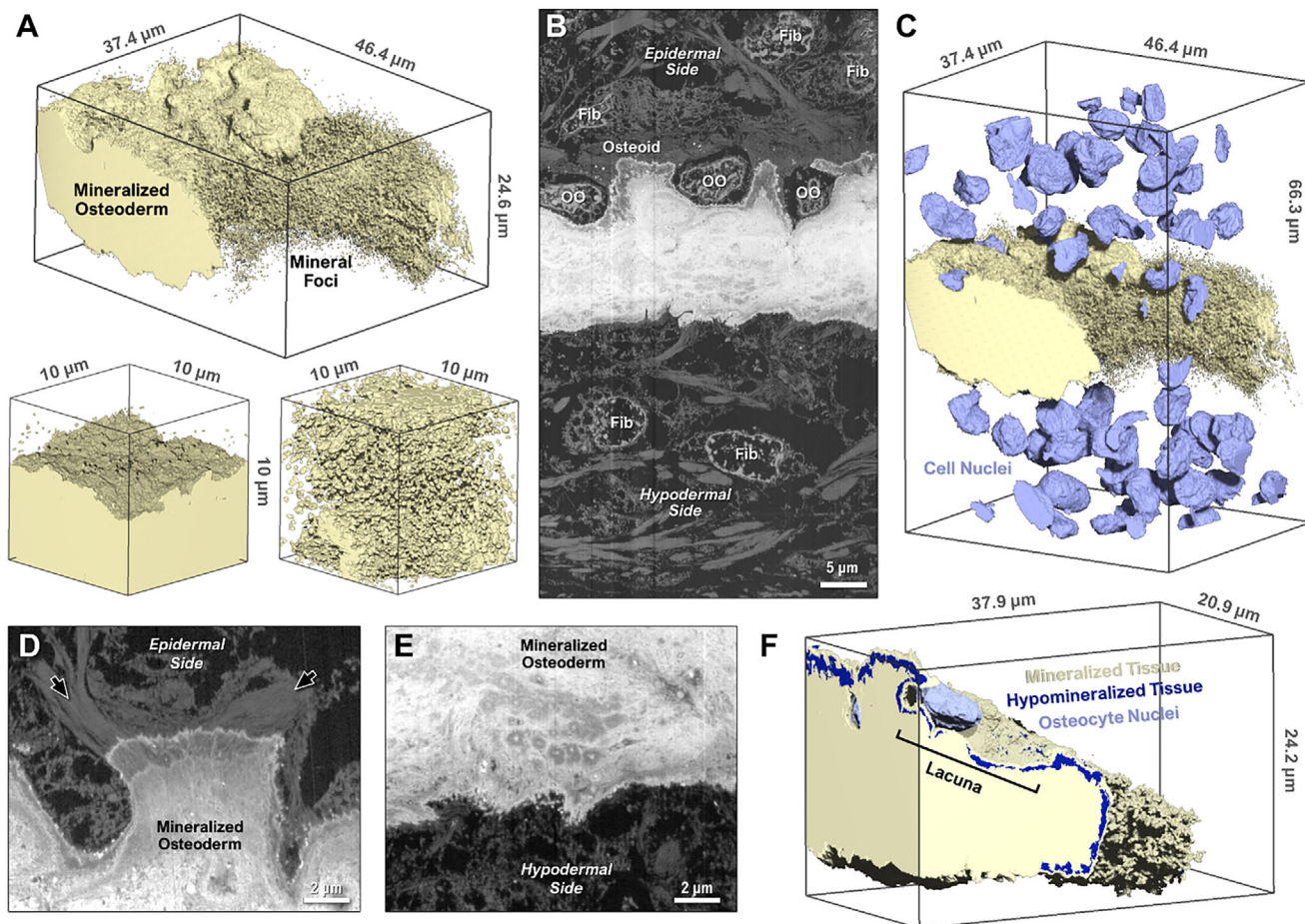


FIGURE 5 | 3D FIB-SEM tomography of mature osteoderm microstructure following osmication. (A) Segmentation of definitively mineralized osteoderm regions within the tomography volume. The volume of interest contains a monolith of fused mineral along with a region of nucleating/growing mineral foci. The surface of the well-mineralized osteoderm has little microtopography, with only sparse incipient foci visible. A subregion in the region of dense mineral foci shows a snapshot of foci coalescence. (B) Reconstructed 2D slice of the FIB-SEM dataset. Fibroblasts are present on both sides of the osteoderm, while cells similar to osteoid osteocytes are only on the epidermal-facing side. (C) 3D reconstruction of cell nuclei (both fibroblasts and osteoid osteocytes) along with osteoderm mineral. (D) Reconstructed 2D region on the epidermal side of the osteoderm, showing dermal collagen inserting into the osteoderm in a Sharpey's fibered orientation. (E) Reconstructed 2D region on the hypodermal side of the osteoderm, showing globular organic inclusions at the base of the osteoderm. (F) Segmented 3D volume showing the hypomineralized band on the epidermal side of the osteoderm. Note its contour around cell lacunae. OO, osteoid osteocyte; Fib, fibroblast. See also: Video S2.

and interior facet of the osteoderm were segmented in 3D using the FIB-SEM tomography data (Figure 7A,B). At the interior side of the osteoderm (extending into the reticular dermis), the dominant collagen orientation runs parallel to the flat surface of the osteoderm – forming an intersecting and interwoven fiber network that wraps circumferentially around the longitudinal axis of the tail. Dermal collagen fibril bundles at the exterior of the osteoderm (extending into the papillary dermis) are equally intersecting and interwoven but have a stronger orientation in the radial direction of the tail. VectoRose analysis [32] of the two segmented areas shows a visual representation of collagen orientation, using segmentation-derived anisotropy vector fields to examine three-dimensional fibril direction. For dermal collagen on the interior side of the osteoderm, there is a high proportion of fibril bundles that lie parallel to the midplane of the osteoderm (represented by a circumscribing band in the spherical histogram, Figure 7C). There are a select few fibrils that have a Sharpey's fibered orientation on the interior surface of the osteoderm (offset by roughly 90° to the osteoderm midplane),

but there are substantially fewer fibrils with this orientation compared to the Sharpey's fibered fibrils on the exterior-facing side of the osteoderm (Figure 7D). The resulting structure in the osteoderm microenvironment provides a strong anchorage in the outermost dermis toward the epidermal layer of the skin through bundled collagen in the papillary dermis, in contrast to a more weakly bound connection to the hypodermal layer of the skin through the reticular dermis (Figure 7E).

3 | Discussion

Here, we have studied the macroarchitecture, microarchitecture, and nanoarchitecture of osteoderms in the tails of the *Acomys cahirinus* spiny mouse, noting major structural deviations from other forms of mineralized tissue (namely those found in bones and teeth). Osteoderms – especially those in mammalian species – therefore appear to be a unique and diverse type of calcified tissue existing in vertebrates that have not yet been properly described

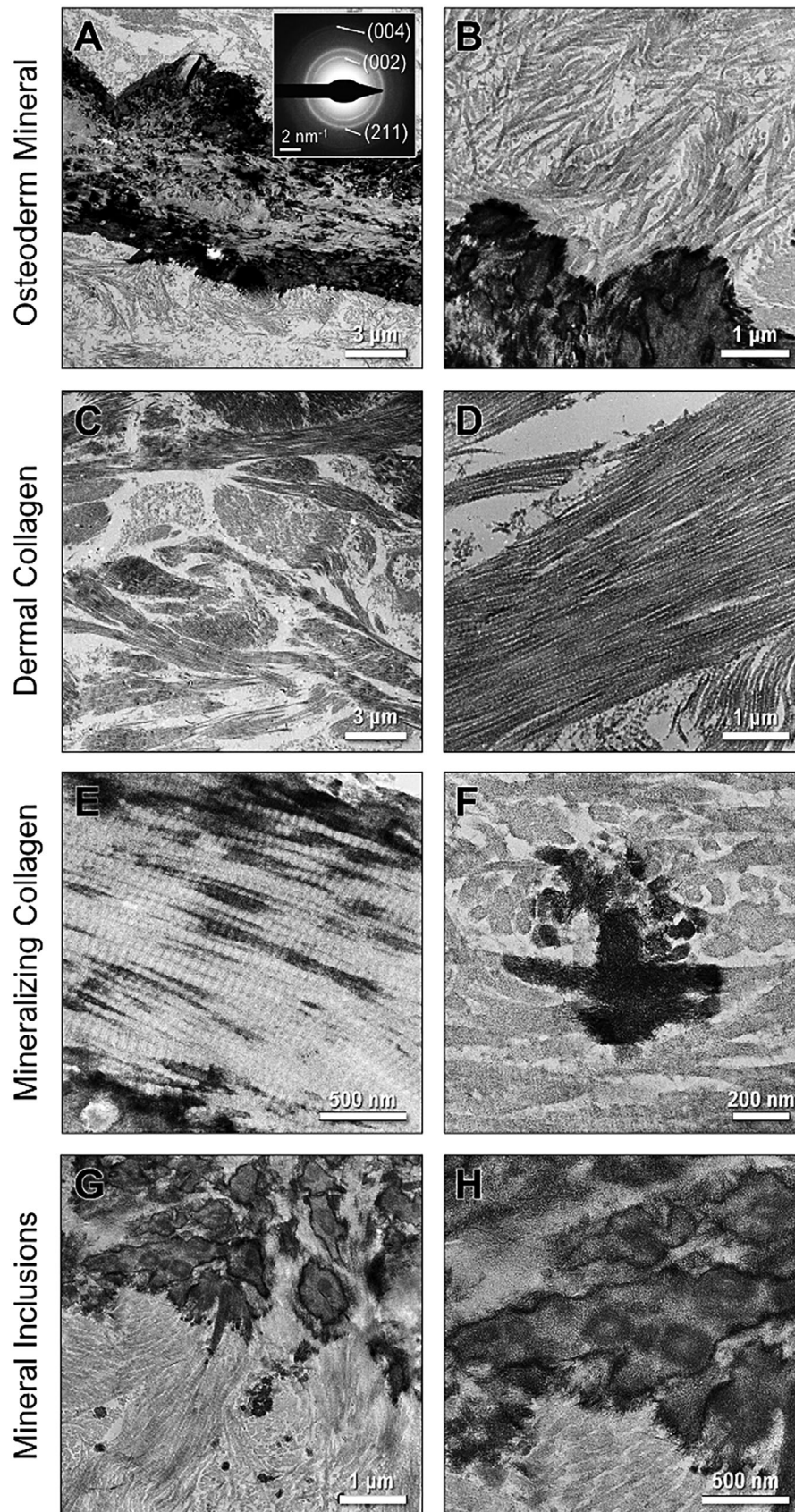


FIGURE 6 | Transmission electron microscopy of mature osteoderm structure following lead citrate and uranyl acetate staining. (A) Imaging of the full osteoderm cross-section. Inset: Selected-area electron diffraction of a nearby area. Major hydroxyapatite reflections of (002), (211), and (004) can be seen. (B) Individual collagen fibrils on the epidermal side of the osteoderm are loosely bundled and inserting in a Sharpey's fiber orientation. (C,D) Tightly bundled dermal collagen arrays well outside of the mineralized osteoderm region, exhibiting a highly unidirectional character. (E) Tightly

in mammals. It follows that the use of bone-based terminology (e.g., osteocyte) may not always be fitting to describe osteoderms but can be used for consistency and comparisons with existing osteoderm literature.

Descriptions of autotomy in these spiny mice (by which they can shed their tails in instances of environmental stress or predation) define an anatomical plane along the longitudinal axis of the tail where skin tears can initiate and propagate [20]. Following autotomy, the shed region is described to contain only the epidermis and outermost layer of the dermis [20] – layers that are remarkably consistent with the positioning of tail osteoderms observed here. This evidence suggests a tenuous connection between osteoderm anatomy and the autotomic fracture plane, implying that osteoderms may also play a role in facilitating autotomy and thereby help the animal survive attacks from predators.

One recurring observation in previous osteoderm studies is the inconsistency with which they are patterned, both across a single species and within a single organism [9]. As an example of heterogeneity within single organisms, the carapace of the *Dasypus armadillo* can be categorized into distinct components depending on osteoderm structure. Osteoderms in the head shield, pectoral shield, and pelvic shield all take on a polygonal shape, while osteoderms in the banded shield and tail shield are concentrically banded with a clear rectangular morphology [18]. Osteoderms in spiny mice, however, display a uniquely homogeneous macroarchitecture with uniformity in osteoderm morphology that is rarely found in other osteoderm-containing species. Macroscopically, osteoderms in the spiny mouse tail have some similarities to each of the specific shield regions in the armadillo. In terms of gross morphology, the spiny mouse osteoderms share the same polygonal morphology found in subregions of the armadillo carapace (head, pelvis, and pectoris), but their organization into well-defined concentric rings is reminiscent of the antiposed regions (tail and banded regions) of the armadillo. Despite this banded, ring-like appearance in the spiny mouse tail, the spacing between osteoderms is relatively consistent in all directions – without the large gaps (as visible in Vickaryous et al. [18]) that interpose each concentric ring in the embryonic armadillo. The staggered A-B-A-B-A-B patterning may also be beneficial for avoiding the formation of extended seams of soft tissue by forming a brickwork-like pattern. Such patterning is reminiscent of nature's biological reaction-diffusion products (reviewed in [33]).

The intrinsic heterogeneity found within reptilian osteoderms can complicate any broad structural comparison between reptilian osteoderms and spiny mouse osteoderms. Microstructural motifs similar to that of woven bone, lamellar bone, Sharpey-fibered bone, cortical bone, and cancellous bone all exist in the osteoderms of various reptilian species amidst a wide variety of macroarchitectures [9]. However, the imbricated, rectangular,

and overlapping macrostructure found in the *Acomys cahirinus* osteoderm can sometimes be found in Reptilia – particularly in the dermis of scincids and anguids – but less is known about the microstructural organization in these species, making direct comparisons challenging.

By necessity, the microstructure of *Acomys cahirinus* osteoderms should also allow for tail growth over the lifespan of the animal. The initial separation between osteoderms and lack of rigid connections between them is important for accommodating this tail expansion or any other allometric change as the animal grows and matures. Much like suture locations in a growing skull, the spaces between osteoderms can freely widen or narrow over the lifespan of the animal with minimal mechanical constraint from neighboring osteoderms – meaning that they can be positioned independently of other osteoderms in the tail. Osteoderms can therefore maintain a relatively constant separation during tail growth in a way that would not be feasible for a fused or connected mass of mineral. Apart from osteoderms, skeletal growth in mice of this age typically proceed by endochondral or intramembranous ossification mechanisms that seemingly differ from that of osteoderm biomineralization. The ultrastructural aspects of these other tissues are characterized elsewhere [34], but, like osteoderms, the mineralization processes occurring in the cranium and elsewhere are active and ongoing in neonatal mice and the extent of mineralization depends on the precise anatomical location. For instance, previous work has noted that osteoderms only start to appear in the distal third of spiny mice tails after six weeks of age [16], at which time skeletal development is typically quite advanced in other mice species.

On the opposite side of the taxonomic hierarchy, there can exist a superficial layer (100–200 μm thick) of hypermineralized, hypocellularized, and avascular tissue [9] that serves as a capping layer on reptilian osteoderms – often termed osteodermine [35] or capping tissue. This layer is often described as enamel-like because of its mineral-rich and organic-deficient nature – drawing similarities to the outer layer of a tooth. In the young *Acomys cahirinus* spiny mice studied here, we note the absence of a capping tissue, and in fact, note that one external layer of the osteoderm actually appears hypomineralized relative to the bulk of the osteoderm – perhaps since it is in an active state of growth. The lack of capping tissue in spiny mice is consistent with other mammalian osteoderms (namely ground sloths and armadillos [36]), but it should be noted that inconsistencies exist in osteodermine occurrence among species with relative closeness in taxonomic classification [37].

In another analogy to tooth structure, the Sharpey's fiber orientation and disordered network of collagen in the *Acomys cahirinus* osteoderm resemble select regions in the complex structure of tooth cementum [26, 38, 39], where mechanical anchorage and functional loading of teeth in the jaw bones

bundled dermal collagen residing inside of the osteoderm. Mineral platelets grow between, along, and likely within individual collagen fibrils in the overall collagen direction. (F) Mineralization of the less-ordered osteoid collagen on the epidermal side of the osteoderm. Mineral appears between fibrils and densely permeates individual collagen fibrils. (G,H) Globular inclusions present in the hypodermal facet of the mineralized or mineralizing osteoderm. Mineralization initiates in discrete island-like foci, each containing between 1–20 globules of unknown origin.

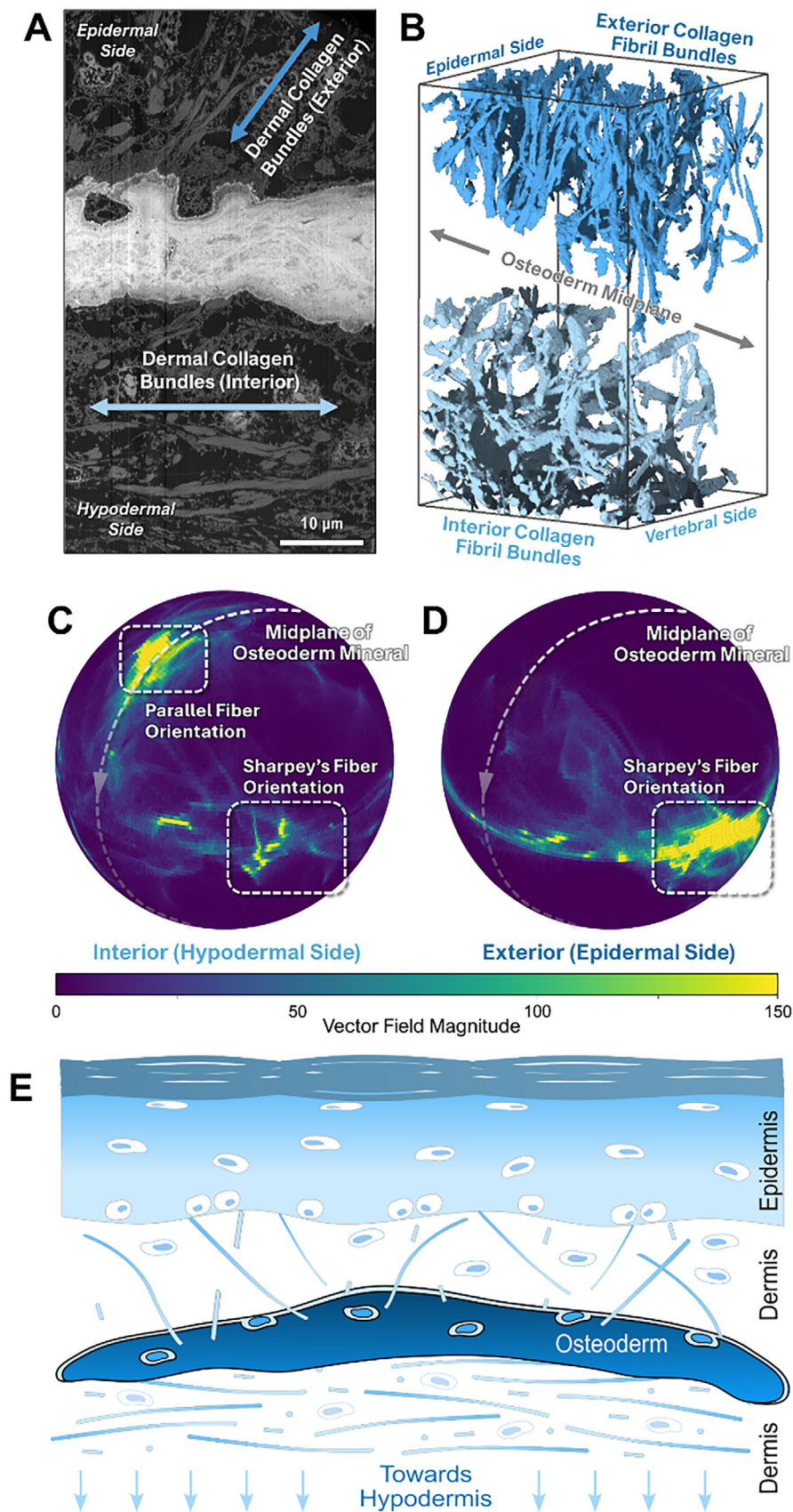


FIGURE 7 | Mapping of Sharpey's fiber orientation in dermal collagen surrounding the mature osteoderm. (A) Reconstructed 2D slice from FIB-SEM tomography. Dermal collagen bundles can be seen in Sharpey's fiber orientation inserting into the osteoderm (epidermal side) and parallel to the osteoderm (hypodermal side). (B) 3D FIB-SEM tomography segmentation of dermal collagen fibrils in the extracellular matrix surrounding the osteoderm (osteoderm digitally removed). Collagen in the hypodermal side tends to be oriented more perpendicularly to the osteoderm surface.

is supported by perpendicular and oblique fiber arrangements and insertions, collectively forming the periodontal ligament. In the *Acomys cahirinus* mouse, the osteoderm mineral takes on a largely monolithic mineral structure at the microscale – similar to the structure of dentin – rather than having clearly demarcated or sub-ordered microscale building blocks (although organic inclusions can often be found in the osteoderm). Many of the highly upregulated genes in osteoderm formation are likened to those of bone [16], but some of these genes can also be implicated in tooth mineralization processes (such as *Bglap2* [40], *Phex* [41, 42], *Spp1* [43], *Omd* [44], and *Sp7* [45]).

In the osteoderm, the inclusion of coarse dermal collagen bundles alongside a more disordered osteoid collagen makes for a structure that is rarely found in bones and teeth. One controversial analog [46] for this collagen archetype occurs from the process of tooth dentinogenesis, where interodontoblastic fibers (also referred to as von Korff fibers) purportedly have similar collagen bundling and are also surrounded by a semi-disordered collagenous matrix [47–50]. These von Korff fibers permeate through the odontoblast layer into the pulp of the tooth, but their function remains largely unclear. The phenomenon of mineral nucleation in these osteoderms, however, more closely resembles that of primary woven bone formation – with an abundance of isolated mineral foci at the growth front and a large fraction of disordered collagen in the mineralizing extracellular matrix. Osteoderm mineralization appears similar to formative events shown by Deering et al. for bone formation in the skull or periosteal collar but is not reminiscent of predentin mineralization (which instead has a largely monolithic form of mineral accretion and only select mineral foci at the mineralization front) [34].

Ultimately, the final internal structure of the osteoderm resembles neither woven bone nor dentin. It follows that the process of osteoderm formation has select elements that can be construed as being either bone-like or tooth-like, but osteoderm biomineralization is best characterized as its own unique process within the integument.

4 | Conclusions

We show here through 2D and 3D multiscale observation that the *Acomys cahirinus* spiny mouse tail osteoderm takes on a complex hierarchical structure, with mineral foci nucleation occurring in a complex organic matrix that consists of:

- i. Highly ordered dermal fibrillar collagen
- ii. Semi-ordered osteoid fibrillar collagen
- iii. Surface-connected cells
- iv. Globular organic inclusions

Osteoderm nucleation sites are precisely positioned relative to existing hair follicles along the tail, creating a consistent pattern of concentrically oriented osteoderms that grow via accretion of mineral in incipient nanoscale foci. Based on the positioning of vascular canals in developing osteoderms, and on the presence of an epidermal-facing hypomineralized domain at the osteoderm surface, we consider that osteoderms have a tendency to preferentially grow in the distal-to-proximal and hypodermal-to-epidermal directions. Overall, these findings provide insight into the developmental and mechanical pathways by which the tail integument is ascribed its evolutionarily remarkable properties – with potential to facilitate autotomy as an anti-predation mechanism.

5 | Methods

5.1 | Animals

Spiny mice (*Acomys cahirinus*) were obtained from a breeding colony kept at the University of Florida and used in accordance with protocols approved by the Institutional Animal Care and Use Committee (IACUC) (Protocol numbers 20240000036 and 202400000403). All animals used in this study are male, as female mice are preserved for breeding, and all neonatal tails used in this study were dissected at a postnatal day 1 (P1) age.

5.2 | X-Ray Microscopy (Submicron X-Ray Microcomputed Tomography)

Spiny mouse tails for x-ray microscopy were amputated from newborn mice and immersed directly into 70% ethanol. x-ray microscopy was conducted on four tails in 70% ethanol in polypropylene tubes using a Zeiss XRadia Versa 520. Full tail scans were conducted using three separate scans with a voltage of 80 kV, power of 7 W, 0.4X objective lens, 0.45 s exposure time, no filter, and a 22.6 μm voxel size across 2401 projections. The three scans were then stitched in Dragonfly 3D World 2024.1 for visualization and quantitative analysis. High-resolution scans of tail osteoderms (in a well-developed region and in a region with newly formed osteoderms from a single, representative tail) were acquired using a voltage of 80 kV, power of 7 W, 4X objective lens, 2.0 s exposure time, proprietary LE2 filter, and a 2.93 μm voxel size across 2601 projections.

The reconstructed and stitched tail was segmented in Dragonfly 3D World using a 2.5D UNet convolutional neural network trained using manually segmented classes (mineral and background) as a ground truth across six training datasets (each containing seven consecutive slices). The neural network was then applied to the full image stack to segment vertebrae and osteoderms across the full tail. The two higher-resolution scans were also segmented using a 2.5D UNet model, trained with

(C,D) Dermal collagen orientation maps in the 3D FIB-SEM volume plotted using VectoRose. Vectors representing the orientation of hypodermal-facing dermal collagen tend to be highly clustered in orientations parallel to the osteoderm surface, with only a minor component of Sharpey's fibers. Vectors representing the orientation of epidermal-facing dermal collagen are highly clustered in a Sharpey's fiber orientation, where few dermal collagen bundles have an orientation parallel to the osteoderm surface. (E) Schematic representation of dermal collagen bundles relative to the osteoderm. See also: Video S2.

three datasets (each containing seven slices of data) of manually segmented mineral, hair follicle, and background classes. In the context of the spiny mice used for this study, the terms “well-developed” and “mature” osteoderms correspond to those residing in proximal regions of the tail – although these are likely still in a period of growth or remodeling. Angular offset from the vertebral longitudinal axis was measured using the angle between the central axis of the least bounding box of the osteoderm and its nearest vertebra. The osteoderm porosity was measured by doing a three-dimensional infill of inner areas on each osteoderm and Boolean subtracting the original segmentation to extract the combined mineral lacunar and vascular porosity. A size threshold was set to exclude porosity larger than 125 voxels, leaving only the lacunar osteoderm porosity. Quantifiable measures (mineral volume and aspect ratio) were measured using connected component analysis. Statistics in this work are reported as the mean value plus or minus the standard deviation.

5.3 | Histology

A single tail from a spiny mouse was amputated from a newborn P1 animal and fixed for ~12 h in 4% paraformaldehyde plus 5% glutaraldehyde in 0.1 M sodium cacodylate buffer, pH 7.3. The tail was then transversely sectioned into segments with a length of ~2 mm before osmication for 1.5 h in 1% OsO₄ plus 1.5% potassium ferrocyanide in dark conditions and at 0°C to add contrast for electron microscopy. A portion of tail with well-mineralized osteoderms was then serially dehydrated in acetone and gradually infiltrated with EMbed812 epoxy resin. After several stages of resin infiltration, resin blocks were cured at 60°C for 48 h and block-trimmed to a site of interest. Light microscopy sections with a thickness of 1 µm were cut from the region of interest using ultramicrotomy and subsequently stained with 3% silver nitrate (von Kossa staining reagent) and 1% toluidine blue solutions. Bright-field imaging was conducted using a Leica DMi8 inverted microscope.

5.4 | Focused Ion Beam Scanning Electron Microscopy (FIB-SEM)

Using an epoxy-infiltrated specimen with well-developed osteoderms, a single tail specimen was first sputter-coated with 5 nm of platinum to add conductivity to reduce charging of the sample. Focused ion beam scanning electron microscopy (FIB-SEM) tomography was performed using a Thermo Scientific Helios Hydra 5 microscope using an inductively coupled oxygen plasma source for all milling stages in tomography and site preparation. A platinum capping layer with a thickness of 2 µm was deposited in the microscope at the site of interest using the gas injection system and ion beam at 30 kV and 5.6 nA. The longitudinal section of the tail was exposed using a rectangular cross-section (30 kV, 0.19 µA) prior to finer cleaning cross-sections (30 kV, 45 nA and 30 kV, 5.6 nA). Trenches were milled on either side of the region to accommodate milled material using an accelerating voltage of 30 kV and current of 45 nA. A fiducial marker was placed beside the region using an ‘X’-shape (30 kV, 1.7 nA, depth of 2 µm) milled into a platinum deposit (30 kV, 0.61 nA, height of 2 µm).

Serial sectioning was performed over four days to acquire the tomogram (2093 images). During tomography acquisition, slices were milled using a 30 kV accelerating voltage, 5.6 nA beam current, milling depth of 60 µm, and slice thickness of 18 nm. Imaging was conducted using backscatter electron signal from the through-lens detector in immersion mode using dynamic focus. Images were acquired using a 2 kV electron beam, probe current of 6.4 nA, 10 µs dwell time, and a 3766 × 2512 camera size (18.0 nm pixel size). Stigmation and focus were automatically adjusted every 10 slices, with contrast/brightness adjustments occurring every five slices. Energy dispersive x-ray spectroscopy (EDS) was performed post-acquisition using an accelerating voltage of 5 kV, probe current of 1.6 nA, and dwell time of 10 µs.

The image stack was aligned in Dragonfly 3D World using both Mutual Information (1% initial step, 0.01% final step) and SSD (1% initial step, 0.01% final step) translation steps. Five slices from the FIB-SEM tomography data were manually segmented into mineral, cell nuclei, and unmineralized extracellular matrix (ECM) classes to train a 2.5D UNet convolutional neural network for image segmentation in Dragonfly 3D World. The trained neural network was applied to the full dataset to segment each of these classes in full. The segmented mineral phase was used as a training/segmentation mask to differentiate the hypomineralized boundary layer of the osteoderm from the bulk osteoderm mineral. A separate 2.5D UNet model was trained to differentiate these components using five consecutive slices of ground truth. The unmineralized ECM region of interest was similarly used as both a training and segmentation mask for a third 2.5D UNet model – trained using two small datasets (each with five slices) to differentiate dermal collagen bundles from non-collagenous features and background in the ECM.

Segmented dermal collagen fibrils were converted to an anisotropy vector field with a preliminary hole-closing step of 0.15 µm, 3× mesh smoothing, a vector field radius of influence of 5 µm, and a vector spacing of 0.36 µm (producing 969 091 and 959 939 non-zero vectors for the interior and exterior dermal collagen fibrils, respectively). Vector fields were exported from Dragonfly 3D World and imported into VectoRose [32] for spherical histogram plotting.

5.5 | Transmission Electron Microscopy

Sections with a thickness of 150 nm were cut from a single aldehyde-fixed and epoxy-embedded tail segment using ultramicrotomy and stained using uranyl acetate and lead citrate solutions to add additional contrast to organic phases. Bright field TEM was conducted in a Thermo Scientific Talos F200X microscope using a voltage of 200 kV. Selected-area electron diffraction was conducted on a separate, unstained section of the osteoderm.

Author Contributions

J. Deering: conceptualization, methodology, formal analysis, investigation, writing – original draft, visualization. **M. Maden:** methodology,

investigation, writing – review and editing, supervision, funding acquisition. **M.D. McKee:** conceptualization, investigation, writing – review and editing, supervision, funding acquisition.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: sml74061-sup-0001-SuppMat.docx.

Supporting File 2: sml74061-sup-0002-VideoS1.mp4.

Supporting File 3: sml74061-sup-0003-VideoS2.mp4.